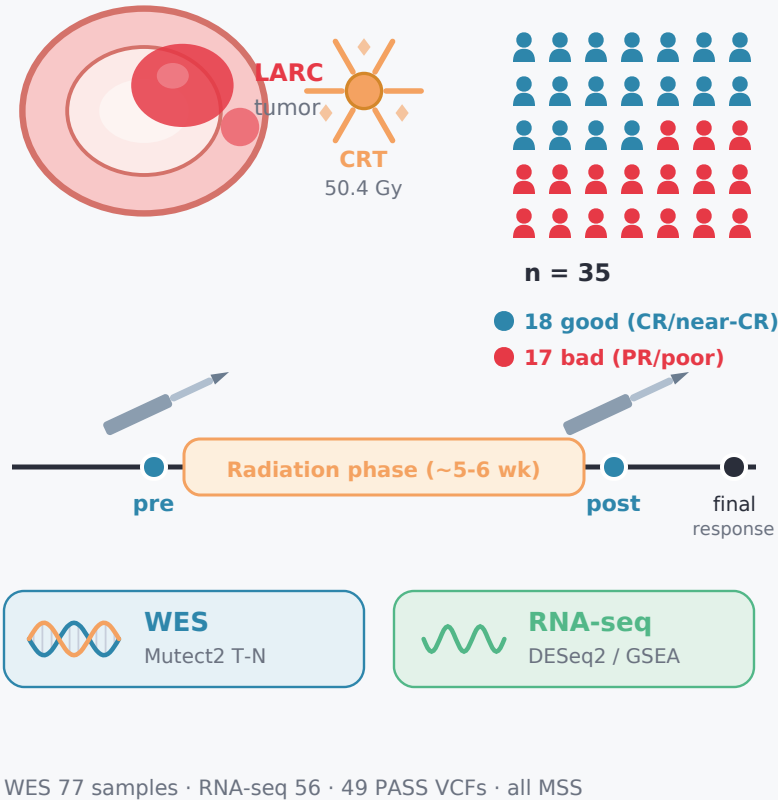


Radiation-Phase Molecular Response Predicts TNT Outcome in MSS LARC

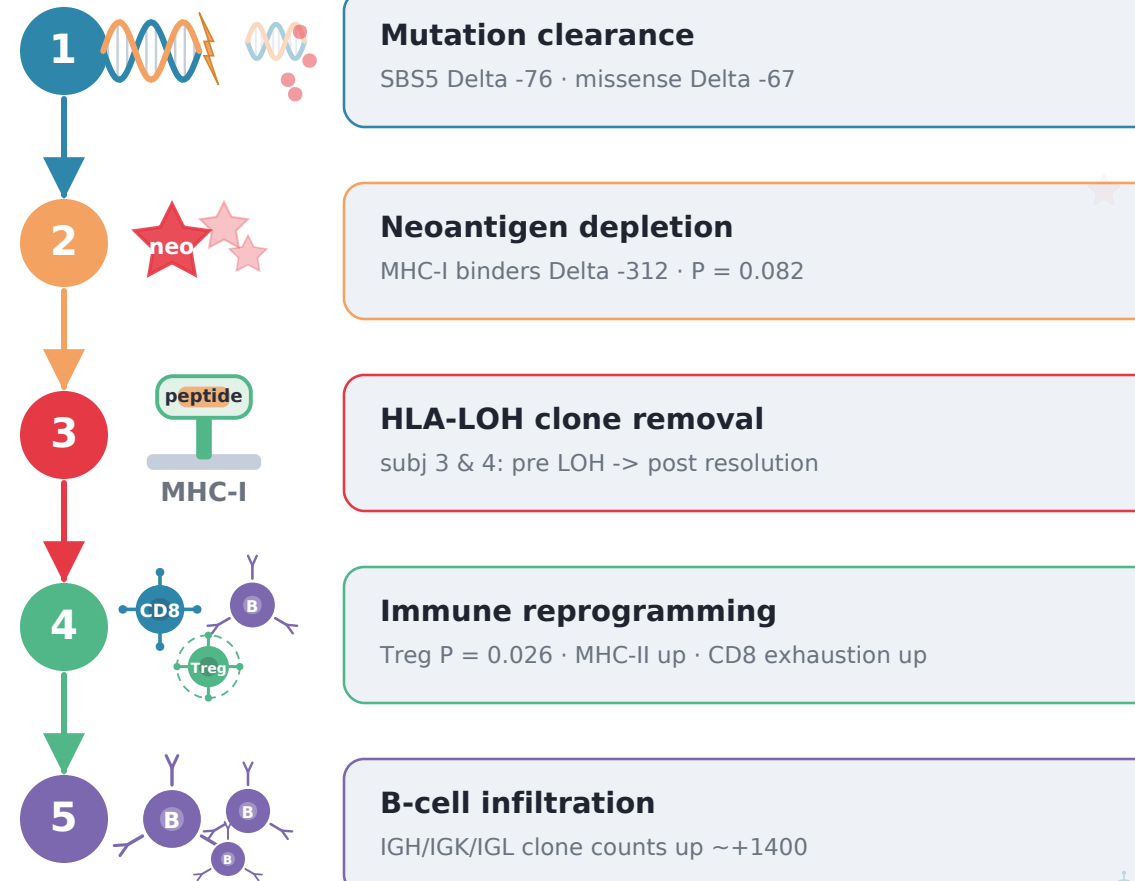
Integrated WES + RNA-seq of 35 locally advanced rectal cancers (pre / post CRT biopsies)

Cohort & Radiation-Phase Sampling



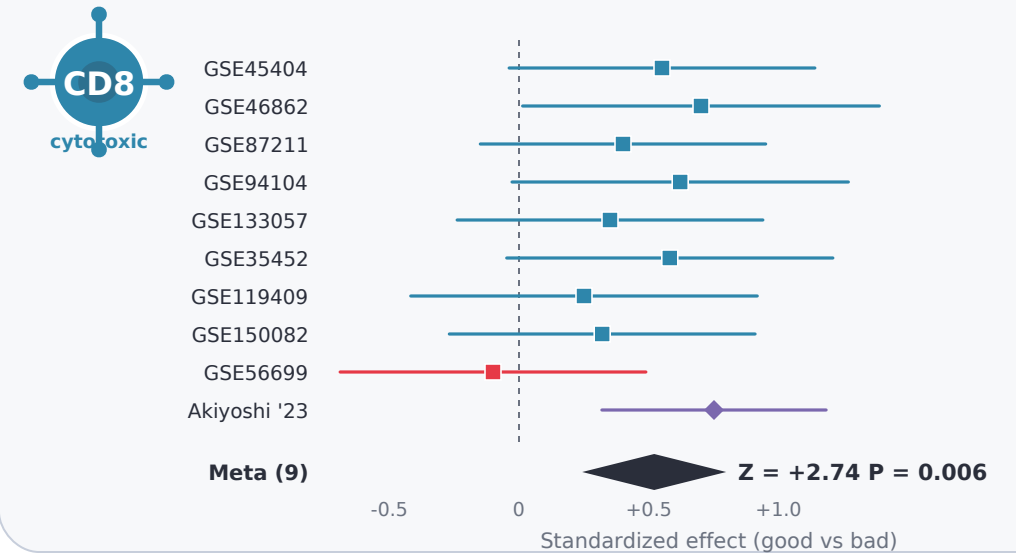
Good-Responder Cascade (pre -> post CRT)

radiation induces sequential tumor clearance & immune reprogramming



External CD8-Cytotoxic Axis Validation

9 nCRT rectal cohorts (N = 721) + Akiyoshi 2023 (N = 298)



Mid-Treatment Decision Window

actionable radiation-phase molecular biomarker

Post-CRT biopsy molecular profile

Responder profile
-> de-escalate / watch-and-wait
organ preservation

Non-responder profile
-> intensify consolidation
FOLFOX / trial enrollment

Take-home: Radiation-phase molecular response -- mutation/neoantigen clearance + reproducible CD8-cytotoxic axis -- defines a mid-treatment biomarker window to personalise consolidation chemotherapy in MSS locally advanced rectal cancer.